

Show understanding of the need for input, output, primary memory and secondary (including removable) storage

9618_w25_qp_13_sc_7.d

QUESTION

(d) Describe **two** reasons why the server that stores the videos uses magnetic hard disks instead of solid state (flash) memory.

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[4]

MARK SCHEME

7(d)	1 mark per reason, 1 mark for corresponding justification x2 max 4 marks • Lower cost per unit of storage • ... so the high capacity of storage required for large number of video files will be less costly • A large number of read/write operations are being performed continuously • ... and magnetic storage is likely to have a longer life span compared with solid state	4
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QUESTION

(ii) The two computers have different amounts of RAM.
Explain how different amounts of RAM affect the performance of a computer.

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..... [3]

MARK SCHEME

6(a)(ii)	1 mark each to max 3 • More RAM means more currently running data and instructions can be stored • without needing to use virtual memory • without having to fetch the data from secondary storage first • which has a slower access time • Less latency / delay waiting for instructions / data	3
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QUESTION

9618/11/M/J/24

(c) The video doorbell has both primary memory and secondary storage.
(i) Identify **two** items of data that the video doorbell will store in primary memory.

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..... [2]

MARK SCHEME

2(c)(i)	1 mark each to max 2 : • Current reading/data from motion sensor • Current/recent video • Instructions being executed • Start-up/BIOS/boot-up instructions	2
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9618_s23_qp_12_sc_5.b

QUESTION

- (b) Describe **two** ways the hardware of a laptop can be upgraded to improve the performance **and** explain how each upgrade improves the performance.

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- 2
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[4]

MARK SCHEME

5(b)	1 mark for description; 1 mark for corresponding explanation Examples • Increase number of cores • Each core can independently carry out a process at the same time // so that more instructions are performed in parallel • Increase RAM capacity • ... allowing more applications to reside in memory at the same time, saving disk access times • Increase cache memory • More data can be stored in fast access so less time is spent accessing from RAM • Increase clock speed • More Fetch-Decode-Execute (FDE) cycles can run each second / per unit time	4
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9618_s23_qp_11_sc_4.b

QUESTION

(b) The central computer stores the video files on secondary storage.

Describe **two** reasons why magnetic storage is more appropriate than solid state storage for this computer.

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- 2
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[4]

MARK SCHEME

4(b)	1 mark for reason, 1mark for application/justification • The computer will have a large number of read/write operations because it is working all the time • .. magnetic storage has more longevity • Magnetic storage costs less per storage unit • ... videos are large files and therefore very large storage capacity is required	4
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Show understanding of embedded systems

9618_w24_qp_12_sc_2.a

QUESTION

2 Embedded systems are used in many electronic devices.

(a) Describe the drawbacks of embedded systems.

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MARK SCHEME

2(a)	1 mark for each bullet point (max 3) e.g. • It is difficult to change / update the firmware by the user • Errors cannot be fixed easily // Troubleshooting / fault-finding / repairing is a specialist task / expensive • Functionality cannot be changed/extended easily // Cannot be easily adapted for another task • Faulty / outdated devices are often thrown away rather than repaired • ... leading to e-waste	3
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QUESTION

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



- (a) The video doorbell can be considered an example of an embedded system.

Identify **two** characteristics of the doorbell that suggest it is an embedded system.

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[2]

MARK SCHEME

2(a)	1 mark each to max 2: • The doorbell only performs the specific tasks of motion detection/video recording/doorbell ringing • The motion sensor and digital camera are built into the doorbell • The CPU/memory/storage/software are all dedicated to this task only • Only a dedicated microprocessor is required due to the limited processing requirements	2
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QUESTION

(c) The factory has many different machines with embedded systems.

(i) Identify **two** features of embedded systems.

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[2]

MARK SCHEME

1(c)(i)	1 mark for each feature (max 2) • Dedicated to a single task // limited number of functions • Built into a larger system // integrated into a larger system • An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	2
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QUESTION

(ii) Identify **one** drawback of embedded systems.

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MARK SCHEME

1(c)(ii)	1 mark for each bullet point (max 1) • Difficult to change / update the firmware by the user // Difficult to upgrade devices to take advantage of new technology • Cannot be easily adapted for another task • Troubleshooting faults/upgrading is a specialist task • Devices are often thrown away rather than repaired as difficult to upgrade or repair	1
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QUESTION

- (b) Give **one** example of an embedded system **and** explain why it is an example of an embedded system.

Example

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Explanation

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[3]

MARK SCHEME

9(b)	One mark for each reason to max 3 to match the example given • Dedicated to one task applied to example • Does not require much processing power applied to example • Built into a larger system applied to example • Contains firmware that cannot be easily updated applied to example • The system does not have its own operating system • An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	3
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QUESTION

- (ii) The car alarm is an example of an embedded system.

Describe the characteristics of an embedded system.

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MARK SCHEME

10(b)(ii)	1 mark for each bullet point (max 3): - the embedded system is built into / integrated (into the car alarm) - combination of hardware and software designed for a specific function - must have a processor, memory and input / output - The system is not easily changed/updated by the car owner	3
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9618_w22_qp_12_sc_3.c

QUESTION

(c) Embedded systems contain Read Only Memory (ROM) and Random Access Memory (RAM).

Explain the reasons why ROM is used in an embedded system.

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MARK SCHEME

3(c)	1 mark for each bullet point: - to store data that does not change - data must be stored even when device is without power - to store boot up instructions / system software / firmware / BIOS	2
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9618_w22_qp_11_sc_9.a

QUESTION

9 Many modern televisions are examples of embedded systems.

(a) Explain why these televisions are embedded systems.

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MARK SCHEME

9(a)	1 mark for each bullet point (max 2): - the embedded system is built into / integrated into the TV - combination of hardware and software designed for a specific function - The system is not easily changed/updated by the TV owner	2
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9618_w22_qp_11_sc_9.b

QUESTION

(b) Embedded systems use Electrically Erasable Programmable ROM (EEPROM).

Describe **one** benefit of using EEPROMs in an embedded system.

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..... [2]

MARK SCHEME

9(b)	1 mark for each benefit and 1 mark for corresponding expansion (max 2). For example: - no additional equipment is needed to change ... enables firmware updates by non-technical users - can be erased and reprogrammed several times ... so firmware can be updated ... can erase a particular byte or the whole EEPROM - possible to reprogram / update ... without removing it from the device	2
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QUESTION

5 Kiara has a washing machine and a refrigerator.

(a) She has an embedded system in her washing machine.

Describe what is meant by an **embedded system**, using the washing machine as an example.

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MARK SCHEME

5(a)	1 mark per bullet point to max 2 • Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task • Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
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Describe the principal operations of hardware devices

9618_w25_qp_13_sc_7.d

QUESTION

(d) Describe **two** reasons why the server that stores the videos uses magnetic hard disks instead of solid state (flash) memory.

1

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2

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[4]

MARK SCHEME

7(d)	1 mark per reason, 1 mark for corresponding justification x2 max 4 marks • Lower cost per unit of storage • ... so the high capacity of storage required for large number of video files will be less costly • A large number of read/write operations are being performed continuously • ... and magnetic storage is likely to have a longer life span compared with solid state	4
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QUESTION

(b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.

(i) Explain how a touchscreen converts the point of touch to a selection from the menu.

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MARK SCHEME

8(b)(i)	1 mark per bullet point, max 4 marks Methods to determine point of touch (Max 2) • Resistive // two layers of material make contact and complete a circuit • Capacitive // contact with the screen creates a change in charge • Infrared // the beams are broken by the pointing device • Optical imaging // a shadow is created when the surface is touched • Acoustic pulse // the acoustic wave is absorbed by the pointing device • Selection from the menu (Max 2) • The point of touch is used to determine the x and y coordinates • The menu item corresponding to the coordinate position is recognised and the menu item is added to the order	4
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QUESTION

10 An architect needs a model of a building.

(a) Explain how the model will be printed using a 3D printer.

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..... [3]

MARK SCHEME

10(a)	1 mark per bullet point, max 3 marks Generic mark points: (max 3) • Additive manufacturing • Uses a digital file created from 3D modelling software or Computer Aided Design (CAD) software • The printer builds up the model one layer at a time • The process is repeated for each layer • Some materials may need time to cool and set or UV curing using LEDs for photopolymer resins Specific mark points: (max 1) e.g. • Fused Deposition Modelling (FDM) // Material is heated and pushed through nozzle / extruder • Stereolithography (SLA) // Photosensitive liquid resin is exposed to a UV-laser beam • Digital Light Processing (DLP) // Photosensitive liquid resin is exposed to a UV projected image of the layer to cure the resin • Selective Laser Sintering (SLS) // Uses a laser to form objects from powdered material, the material can be plastics or metals	3
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QUESTION

6 A programmer is buying a new computer.

(a) The programmer is considering the following two computers:

Computer 1	Quad-core 2.2GHz processor 16-bit architecture 1 GB Random Access Memory (RAM) 500 GB magnetic hard disk
Computer 2	Dual-core 3.8GHz processor 32-bit architecture 2 GB RAM 500 GB solid state drive (SSD)

(i) Computer 1 has a magnetic hard disk.

Complete the description of the principal operation of a magnetic hard disk by writing the missing words.

The magnetic hard disk has one or more that can be magnetised. These are mounted on a and rotate at high speed.

A is moved across the surface on an arm.

When data is read, the changes in the produce a change in the electric current.

[4]

MARK SCHEME

6(a)(i)	1 mark for each correctly completed space • platters • spindle • read/write head • magnetic field The magnetic hard disk has one or more platters that can be magnetised. These are mounted on a spindle and rotate at high speed. A read/write head is moved across the surface on an arm. When data is read the changes in the magnetic field produce a change in the electric current.	4
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QUESTION

(d) The computer system is used to store data received from a temperature sensor every five seconds. The data is stored on an optical disc using an optical disc reader/writer.

(i) Describe the principal operation of an optical disc reader/writer.

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..... [4]

MARK SCHEME

3(d)(i)	1 mark for each bullet point (max 4) • The disc is spun at high speed • A laser is shone onto the disc to read / write • ... using optical head to move it into position • ... it follows the spiral track from the centre outwards • When writing the laser burns pits to represent the data • When reading the laser reflects from pits and lands • The reflection from a pit and a land is different • ... the differences are interpreted as 1 or 0	4
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QUESTION

(d) The robot includes a touchscreen for the customer to make their payment.

Describe the principal operation of a touchscreen.

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..... [4]

MARK SCHEME

7(d)	1 mark each to max 4: • Resistive: The space between the conductive layers is removed/the layers touch and a circuit is completed • Capacitive: The electrical charge changes where the user pressed • The point of contact is identified • ... from the change in electrical field • The software/microprocessor calculates the coordinates	4
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QUESTION

- 2 A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

(a) Complete the description of the principal operation of a VR headset.

A headset can have one or two that output the image to the user. The headset has speakers that output surround sound to give a realistic experience.

The user's head movements are detected using a sensor.

This sensor is a The data is transmitted to a microprocessor that analyses the data to identify the of movement. Some headsets use that record the user's eye movements for analysis.

[4]

MARK SCHEME

2(a)	1 mark for each correctly completed statement: • (LCD) displays/screens/lenses • gyroscope/accelerometer • direction/speed • digital cameras A headset can have one or two (LCD) displays/screens/lenses that output the image to the user. The headset has speakers that output surround sound to give a realistic experience. The user's head movements are detected using a sensor. This sensor is a gyroscope/accelerometer. The data is transmitted to a microprocessor that analyses the data to identify the direction/speed of movement. Some headsets use digital cameras that record the user's eye movements for analysis.	4
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QUESTION

- (ii) The video doorbell has a solid state (flash) secondary storage device.

Complete the table by writing the answer or answers to each statement about the principal operation of solid state (flash) memory.

Statement	Answer
the two types of logic gate that can be used to create solid state devices	1 2
the number of transistors contained in each cell	
the type of gate that can retain electrons without power	
the type of gate that allows or stops current from passing through	

[4]

MARK SCHEME

2(c)(ii)	1 mark for each row:	4										
<table><tr><th>Statement</th><th>Answer</th></tr><tr><td>The two types of logic gate that can be used to create solid state devices</td><td>NAND NOR</td></tr><tr><td>The number of transistors contained in each cell</td><td>2</td></tr><tr><td>The type of gate that can retain electrons without power</td><td>floating</td></tr><tr><td>The type of gate that allows or stops current from passing through</td><td>control</td></tr></table>		Statement	Answer	The two types of logic gate that can be used to create solid state devices	NAND NOR	The number of transistors contained in each cell	2	The type of gate that can retain electrons without power	floating	The type of gate that allows or stops current from passing through	control	
Statement	Answer											
The two types of logic gate that can be used to create solid state devices	NAND NOR											
The number of transistors contained in each cell	2											
The type of gate that can retain electrons without power	floating											
The type of gate that allows or stops current from passing through	control											

QUESTION

(c) Describe how the laser printer makes use of a buffer.

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..... [4]

MARK SCHEME

7(c)	1 mark for each bullet point (max 4) • The print instructions and data are sent by the laptop to a buffer (at laptop speed) • The data is transferred from the buffer to the printer (at printer speed) • ...allowing user to continue using the laptop // and allowing processor to continue processing • ...instead of waiting for relatively slower printer • When the buffer is empty an interrupt is sent to the laptop • ...requesting more data	4
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QUESTION

7 (a) Describe the principal operations of a 3D printer.

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..... [3]

MARK SCHEME

7(a)	1 mark for each bullet point (max 3) Generic mark points: • Additive manufacturing • Uses a digital 3D model or a Computer Aided Design (CAD) (file) • Builds up the model one layer at a time • ...starting from the bottom • ...using x, y and z co-ordinates • The material is fused / cured together layer by layer Specific mark points: Fused Deposition Modelling (FDM) • Material is heated and pushed through nozzle / extruder Stereolithography (SLA) • Photosensitive liquid resin is exposed to a UV-laser beam Digital Light Processing (DLP) • Uses liquid plastic resin melted with arc lamps Selective Laser Sintering (SLS) • Uses a laser to form objects from powdered material	3
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9618_w23_qp_11_sc_7.b

QUESTION

(b) Describe the purpose of a temperature sensor within the 3D printer.

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..... [2]

MARK SCHEME

7(b)	1 mark for each bullet point (max 2) • To prevent overheating // ensure material is hot enough • ...by identifying the temperature of the object (being printed) • ...by identifying the temperature of the material being used	2
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QUESTION

3 A mobile telephone is used to record a video.

(a) The mobile telephone has a touchscreen. There are different types of touchscreen.

Complete the description of the principal operation of touchscreens.

A touchscreen has two layers. When the user touches the screen, the layers touch and a is completed.

A touchscreen has several layers. When the top layer is touched, there is a in the electric current.

A microprocessor identifies the of the touch.

[5]

MARK SCHEME

3(a)	1 mark for each term • Resistive • Circuit • Capacitive • Change • Coordinates A resistive touchscreen has two layers. When the user touches the screen, the layers touch and a circuit is completed. A capacitive touchscreen has several layers. When the top layer is touched there is a change/increase/decrease in the electric current. A microprocessor identifies the coordinates of the touch.	5
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QUESTION

(b) The central computer stores the video files on secondary storage.

Describe **two** reasons why magnetic storage is more appropriate than solid state storage for this computer.

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[4]

MARK SCHEME

4(b)	1 mark for reason, 1mark for application/justification • The computer will have a large number of read/write operations because it is working all the time • .. magnetic storage has more longevity • Magnetic storage costs less per storage unit • ... videos are large files and therefore very large storage capacity is required	4
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QUESTION

- 3 Andy likes to play computer games.
- (a) Andy uses several input devices to play the games. These include a keyboard and a microphone.

Describe the principal operation of a microphone.

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..... [3]

MARK SCHEME

3(a)	1 mark for each bullet point to max 3 • The microphone has a diaphragm / ribbon • The incoming sound waves cause vibrations of the diaphragm • ... causing a coil to move past a magnet // causing a magnet to move past a coil (dynamic microphone) // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) • An electrical signal is produced	3
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Show understanding of the use of buffers

9618_w25_qp_11_sc_10.b

QUESTION

(b) State why a 3D printer needs a buffer.

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..... [1]

MARK SCHEME

10(b)	1 mark • To free up the processor to carry out other tasks as the rate data is received by the printer is different from the rate at which it can be processed • To manage the mismatch in speed between the processor and the peripheral / 3D printer	1
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9618_w24_qp_13_sc_2.b

QUESTION

(b) Explain how a memory buffer is used when a computer is transferring data to its magnetic hard disk drive.

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..... [4]

MARK SCHEME

2(b)	1 mark for each bullet point (max 4) - The computer and the hard disk drive transmit and receive at different speeds // The computer transfers data faster than the HDD can receive - The buffer is used for temporary storage ... so that the computer can transfer data to the buffer at the higher speed ... and is not held up waiting for data to transfer ... and so that data is transferred to hard disk drive from the buffer at the slower rate	4
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9618_w24_qp_11_sc_3.d.ii

QUESTION

(ii) The computer uses a buffer when writing data to the optical disc.

Explain the use of a buffer when writing data to the optical disc.

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..... [3]

MARK SCHEME

3(d)(ii)	1 mark for each bullet point (max 3) - The computer and optical disc reader/writer send and receive at different speeds - A buffer allows temporary storage of the data ... so that the computer can transfer data to the buffer at the higher speed ... and is not held up waiting for data to transfer // so the computer can carry on with other tasks ... so the optical disc reader/writer is not overloaded ... and so that data is transferred to optical disc reader/writer from the buffer at the slower rate	3
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QUESTION

(b) The computer uses a buffer when transmitting data to the VR headset.

Explain how a buffer is used when data is transmitted between the computer and the VR headset.

[3]

MARK SCHEME

2(b)	1 mark each to max 3:
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QUESTION

(iii) The video doorbell uses a buffer.

Describe how the video doorbell will use the buffer.

[2]

MARK SCHEME

2(c)(iii)	1 mark each to max 2: • Captured video is transmitted to buffer • ... video is transmitted from buffer to smartphone • Store recent data in a buffer for the user to rewind • ... instead of storing everything in secondary storage • Store readings from motion sensor • ... until the microprocessor can process them • Store video from digital camera • ... before moving it to secondary storage	2
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9618_w23_qp_13_sc_7.c

QUESTION

(c) Describe how the laser printer makes use of a buffer.

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..... [4]

MARK SCHEME

7(c)	1 mark for each bullet point (max 4) • The print instructions and data are sent by the laptop to a buffer (at laptop speed) • The data is transferred from the buffer to the printer (at printer speed) • ...allowing user to continue using the laptop // and allowing processor to continue processing • ...instead of waiting for relatively slower printer • When the buffer is empty an interrupt is sent to the laptop • ...requesting more data	4
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QUESTION

- (ii) Describe the roles of the address bus, the data bus **and** buffers in the process of writing data to the optical disc reader/writer.

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..... [3]

MARK SCHEME

8(c)(ii)	1 mark for each component (max 3) Buffers - A buffer temporarily holds data until it is ready to be transmitted to the device Address Bus - The address of the data to be written to the device (in RAM) is carried on the address bus Data Bus - All data to be written to the device / buffer is carried on the data bus	3
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QUESTION

- (c) A computer uses a buffer when playing the audio message.

Explain the purpose of a buffer in a computer system using **one other** example.

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MARK SCHEME

1(c)	1 mark for purpose (max 2): Purpose: • to act as temporary storage // to store (downloaded) data • ... before it is used by the receiving device • ... to allow processes / devices to operate at different speeds // independently of each other 1 mark for each example (max 1): Examples: • printer buffer used when data is transferred from a computer to a printer • video buffer when streaming videos • keyboard buffer when performing data entry	3
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Explain the differences between Random Access Memory (RAM) and Read Only Memory (ROM)

9618_s23_qp_11_sc_4.a.i

QUESTION

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (a) This computer has both Read Only Memory (ROM) and Random Access Memory (RAM).

- (i) Describe the contents of the ROM in the central computer.

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..... [2]

MARK SCHEME

4(a)(i)	1 mark each to max 2 • Stores the bootstrap program // start-up instructions for the central computer // BIOS • Stores the start-up instructions for the CCTV system/cameras // firmware for CCTV • Stores the kernel of the Operating System // stores parts of the Operating System	2
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QUESTION

(c) Embedded systems contain Read Only Memory (ROM) and Random Access Memory (RAM).

Explain the reasons why ROM is used in an embedded system.

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..... [2]

MARK SCHEME

3(c)	1 mark for each bullet point: - to store data that does not change - data must be stored even when device is without power - to store boot up instructions / system software / firmware / BIOS	2
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QUESTION

(b) The washing machine's embedded system makes use of both Random Access Memory (RAM) and Read Only Memory (ROM).

State the purpose of RAM and ROM within the washing machine's embedded system.

RAM

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ROM

..... [2]

MARK SCHEME

5(b)	1 mark for RAM, 1 mark for ROM RAM: • Store the choices/wash program the user has entered // stores the data read from the sensors // stores the time left in the program // by example ROM: • Store the start-up instructions (for the washing cycles)	2
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Explain the differences between Static RAM (SRAM) and Dynamic RAM (DRAM)

9618_w25_qp_11_sc_6.c

QUESTION

(c) State **three** differences between Dynamic RAM (DRAM) and Static RAM (SRAM).

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3

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[3]

MARK SCHEME

6(c)	1 mark per bullet point, max 3 marks • DRAM requires to be refreshed/charged SRAM does not request a refresh • DRAM stores each bit as charge SRAM uses flip-flop to store each bit • DRAM is less expensive to manufacture SRAM is more expensive to manufacture • DRAM has slower access speeds SRAM has faster access times • DRAM has higher <u>storage/bit/data density</u> SRAM has lower storage/bit/data density • DRAM is used in main memory SRAM is used in cache	3
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QUESTION

2 A memory buffer uses Dynamic RAM (DRAM).

(a) Identify **two** differences between DRAM and Static RAM (SRAM).

Difference 1

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Difference 2

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[2]

MARK SCHEME

2(a)	1 mark for each difference e.g. • DRAM requires constant refreshing, but SRAM does not need to be constantly refreshed • DRAM generally has greater capacities per chip, but SRAM generally has lower capacities per chip • DRAM has slower access times than SRAM	2
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QUESTION

(b) An embedded system has Dynamic RAM (DRAM).

Identify **one** benefit of using DRAM instead of Static RAM (SRAM) in an embedded system.

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MARK SCHEME

2(b)	1 mark from: • DRAM is less expensive to manufacture/purchase than SRAM in the embedded system • DRAM has a higher bit density per chip // more data can be stored per chip	1
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QUESTION

- (c) Identify **two** disadvantages of using Dynamic RAM (DRAM) instead of Static RAM (SRAM) in a computer system.

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[2]

MARK SCHEME

3(c)	1 mark for each bullet point (max 2) e.g. • DRAM requires constant refresh cycles unlike SRAM • DRAM has lower access speed than SRAM	2
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QUESTION

- 7 A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

- (a) Explain why Static RAM is used in the laptop instead of Dynamic RAM.

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..... [2]

MARK SCHEME

7(a)	1 mark for each bullet point (max 2). • Static RAM has faster access time • ...because it does not need to be refreshed • ...used on the CPU for improvement of CPU cache speed • Static RAM has lower	2
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QUESTION

(c) A 3D printer contains 1 GB of Dynamic RAM (DRAM) to store print data.

State **two** advantages of the printer having Dynamic RAM instead of Static RAM (SRAM).

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[2]

MARK SCHEME

7(c)	1 mark for each bullet point (max 2) - Dynamic RAM has lower cost per unit - A fast access speed is not needed - Higher bit density // more data can be stored per chip	2
------	---	----------

QUESTION

(ii) The central computer has Dynamic RAM (DRAM).

Identify **two** advantages of using DRAM instead of Static RAM (SRAM).

1

.....

2

.....

[2]

MARK SCHEME

4(a)(ii)	1 mark each to max 2 - Costs less per unit - Higher storage density - Simple design – uses fewer transistors	2
----------	--	----------

9618_w25_qp_13_sc_4.c

QUESTION

- (c) Give **two** differences between Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM).

Difference 1

.....

.....

Difference 2

.....

.....

[2]

MARK SCHEME

2(c)	1 mark for each bullet point (max 2) • EPROM uses ultraviolet light to erase data whilst EEPROM uses an electrical signal to do this • EPROM has to be removed from the circuit board when changing the data whilst EEPROM remains in the circuit when the data is changed • EPROM erases all the data, EEPROM can erase parts of the data	2
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QUESTION

- (c) The VR headset has Electrically Erasable Programmable Read Only Memory (EEPROM).

Explain the benefits of using EEPROM instead of other types of Read Only Memory (ROM) in the VR headset.

.....

.....

.....

.....

.....

.....

[3]

MARK SCHEME

2(c)	1 mark each to max 3: • EEPROM allows frequent/multiple read/write/erase operations • ... so the headset can take advantage of new features • ... without fully erasing the contents of the firmware in the headset first // can erase a particular byte or the whole EEPROM • ... without removing the chip(s)/firmware from the headset • ... the contents of the firmware in the headset can be changed by the user without technical expertise • Cheaper to manufacture so headset will be cheaper to purchase	3
------	---	----------

9618_w23_qp_13_sc_7.b

QUESTION

(b) Identify **two** reasons for using Electrically Erasable Programmable ROM (EEPROM) in a virtual reality headset.

1

.....

2

.....

[2]

MARK SCHEME

7(b)	1 mark for each bullet point (max 2) • EEPROM allows frequent / multiple read / write / erase operations • ...which means that the contents of the firmware in the headset can be changed easily • ...without fully erasing the contents of the firmware in the headset first	2
------	--	----------

9618_w22_qp_11_sc_9.b

QUESTION

(b) Embedded systems use Electrically Erasable Programmable ROM (EEPROM).

Describe **one** benefit of using EEPROMs in an embedded system.

.....

.....

.....

.....

[2]

MARK SCHEME

9(b)	1 mark for each benefit and 1 mark for corresponding expansion (max 2). For example: • no additional equipment is needed to change • ... enables firmware updates by non-technical users • can be erased and reprogrammed several times • ... so firmware can be updated • ... can erase a particular byte or the whole EEPROM • possible to reprogram / update • ... without removing it from the device	2
------	--	----------

Show an understanding of monitoring and control systems

9618_w25_qp_12_sc_11

QUESTION

- 11 An automated system opens doors when a person is detected within 2 metres. The system closes the doors when there is no longer a person within 2 metres.

Identify whether the automated system is an example of a monitoring system or a control system.

Justify your choice.

Monitoring or Control

Justification

.....

.....

.....

.....

..... [3]

MARK SCHEME

11	1 mark per bullet point, max 3 marks Identification: Control system (no mark) Justification: • The system uses an actuator to open or close the doors • The output / opening the doors allows people to move, which changes the input to the sensor • The system acts autonomously on the feedback from a sensor	3
----	--	----------

QUESTION

9

Describe the differences between a monitoring system and a control system.

[3]

MARK SCHEME

9	1 mark per bullet point, max 3 marks - Monitoring systems do not take any action whereas control systems act autonomously to change the environment if the values are out of a prescribed range - Control systems use actuators monitoring systems do not have any actuators - Control systems make use of feedback monitoring systems do not make use of feedback - The output from a monitoring system does not affect the subsequent input whereas output from a control system affects the next input.	3
---	--	---

QUESTION

(b)

The new system uses sensors to identify the items taken from a shelf.

Identify **one** type of sensor that can be used in this new system.

State how the sensor can be used to identify the items taken from a shelf.

Sensor

Use

[2]

MARK SCHEME

4(b)	1 mark for sensor, 1 mark for corresponding use e.g. <table><tr><th>Sensor</th><th>Use</th></tr><tr><td>Pressure</td><td>Detects when the pressure of an item is removed from a shelf / put back on a shelf</td></tr><tr><td>Infrared</td><td>Detects when the beam is broken for an item removed / added</td></tr></table>	Sensor	Use	Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf	Infrared	Detects when the beam is broken for an item removed / added	2
Sensor	Use							
Pressure	Detects when the pressure of an item is removed from a shelf / put back on a shelf							
Infrared	Detects when the beam is broken for an item removed / added							

9618_w24_qp_13_sc_3

QUESTION

- 3 A car has an automatic braking system.

A sensor is used to measure the distance to objects that are in front of the car.

The car automatically brakes if an object is too close to the front of the car or the distance between the car and the object is decreasing too quickly. The closer the object is to the front of the car, the harder the car brakes so that the car slows down more quickly.

Explain the reasons why the automatic braking system of the car is a control system.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

3	1 mark for each bullet point (max 3) • The system uses feedback • The system acts independently • Input data causes the car to brake • ... braking decreases the/changes the distance to the object • ... this new distance is used to determine whether/how hard the car needs to brake	3
---	--	----------

QUESTION

- 9 A road bridge has a weight limit and a height limit for vehicles. For example, a vehicle must weigh less than 10 000 kg and must have a height of less than 3 m.

The bridge has a warning system. If a vehicle is approaching the bridge and it exceeds one or both limits, a sign displays a warning telling the driver of the vehicle to stop.

- (a) The bridge warning system uses sensors to detect if a vehicle exceeds the limits.

Complete the table by identifying **two** different sensors that could be used by the system **and** describe how each sensor is used by the system.

Sensor	Use in bridge warning system
.....	
.....	
.....	
.....	
.....	
.....	

[2]

MARK SCHEME

9(a)	1 mark for sensor and its use in this system • Infra-red sensor • Measure / check the height of the vehicle • Pressure sensor • Measure / check the weight of the vehicle	2
------	--	----------

QUESTION

- (b) Explain whether the bridge warning system is an example of a monitoring system or of a control system.

.....

.....

.....

..... [2]

MARK SCHEME

9(b)	1 mark for each bullet point (max 2) for a correct justification No marks for the identification of the system Monitoring system • Because there is no use of feedback // the warning sign is only an indicator • ... the output of the turning on of the sign does not affect the input of data from the sensors • The system does not have any actuators	2
------	---	----------

QUESTION

- 5 A security system has both a floodlight (very bright light) and an audio alarm.

The data from multiple sensors is analysed and used to:

- turn on the floodlight
- sound the audio alarm.

Sensors can be used to detect:

- if doors are open
- the external daylight level
- if people are detected within a set distance.

- (a) Complete the table to identify the most appropriate type of sensor for each scenario.

Scenario	Type of sensor
A door is open.	
The external daylight level is below a set amount.	
A person is detected within 2 metres.	

[1]

MARK SCHEME

5(a)	1 mark for all 3 appropriate sensors	1								
<table><tr><th>Scenario</th><th>Type of sensor</th></tr><tr><td>A door is open.</td><td>Pressure sensor // Infra-red</td></tr><tr><td>The external daylight-level is below a set amount.</td><td>Light sensor</td></tr><tr><td>A person is detected within 2 metres.</td><td>Infra-red</td></tr></table>		Scenario	Type of sensor	A door is open.	Pressure sensor // Infra-red	The external daylight-level is below a set amount.	Light sensor	A person is detected within 2 metres.	Infra-red	
Scenario	Type of sensor									
A door is open.	Pressure sensor // Infra-red									
The external daylight-level is below a set amount.	Light sensor									
A person is detected within 2 metres.	Infra-red									

QUESTION

- (c) Explain whether the security system is an example of a monitoring system or a control system.

.....

.....

.....

.....

.....

..... [3]

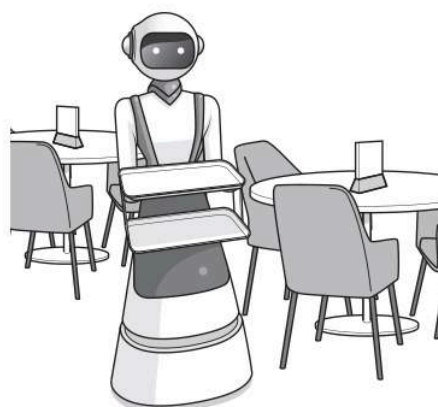
MARK SCHEME

5(c)	1 mark for each bullet point (max 3) for a correct justification No marks for the identification of the system Monitoring system • There is no use of feedback // The light and the alarm are just warnings • The output of the floodlight or the audio alarm does not affect the input of data from the sensors • The system does not have any actuators	3
------	--	----------

QUESTION

13

- 7 Robots are used to serve food and drink to customers at a restaurant.



- (a) A robot navigates through the restaurant to the table it is serving.

Complete the table by identifying **two** sensors that can be included in the robot **and** the purpose of each sensor in the navigation system.

Sensor	Purpose of sensor in navigation system
.....	
.....	
.....	
.....	

[2]

MARK SCHEME

7(a)	1 mark for sensor and matching purpose to max 2 :	2								
	<table><tr><th>Sensor</th><th>Purpose of sensor in navigation system</th></tr><tr><td>Pressure</td><td>To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on</td></tr><tr><td>Infra-red</td><td>To detect if there is an obstacle in the way // to indicate that it has reached the desired table</td></tr><tr><td>Sound</td><td>To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop</td></tr></table>	Sensor	Purpose of sensor in navigation system	Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on	Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table	Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop	
Sensor	Purpose of sensor in navigation system									
Pressure	To detect if a table or other obstacle has been hit // to detect when food is put on/taken off the tray so it can move on									
Infra-red	To detect if there is an obstacle in the way // to indicate that it has reached the desired table									
Sound	To detect if someone is speaking so that it can use AI to decipher the speech and whether the robot is required to stop									

9618_s24_qp_13_sc_7.c

QUESTION

14

- (c) The navigation system can be considered an example of a control system.

Describe how feedback is used in a control system.

.....

.....

.....

..... [2]

MARK SCHEME

7(c)	1 mark each to max 2 : • Feedback ensures that a system operates within set criteria / constraints • ... by enabling system output to affect subsequent system input • ... thus allowing conditions to be automatically adjusted	2
------	---	----------

9618_s24_qp_11_sc_2.b

QUESTION

- (b) State whether the video doorbell is a monitoring system or a control system.
Justify your choice.

Monitoring or control system

Justification

.....

.....

.....

.....

..... [2]

MARK SCHEME

2(b)	No mark for identification of monitoring or control 1 mark each to max 2 for justification: Monitoring: • The turning on of the digital camera does not affect the input to the sensor/button • The transmission of the data/video does not affect the input to the sensor/button • The ringing of the doorbell does not affect the input to the button Control: • Video doorbell does not only store the values from the motion sensor • The data is processed, generating a signal to start the digital camera recording • Button pressed/motion detected causes a signal to be sent over a network to the smartphone	2
------	---	----------

9618_w23_qp_12_sc_1.a

QUESTION

1 A factory makes chocolate bars. The factory uses a conveyor belt that moves the products from one stage of production to the next stage. (a) An automated system counts the number of chocolate bars made at the end of production. The system includes a sensor positioned above the conveyor belt. Identify one appropriate type of sensor that can be used. [1]

MARK SCHEME

1(a)	1 mark for: • infra-red / proximity (sensor)	1
------	--	----------

9618_w23_qp_12_sc_1.b

QUESTION

- (b) A second automated system removes chocolate bars with an incorrect weight from the production line.

Describe the role of an **actuator** in this second system.

.....
.....
.....
..... [2]

MARK SCHEME

1(b)	1 mark for each bullet point (max 2) • Actuator generates a signal / causes an action / converts electrical energy into a mechanical force • ... to push an arm // to open a trap door // to pick up the chocolate bar with the incorrect weight	2
------	--	----------

9618_w23_qp_11_sc_7.b

QUESTION

- (b) Describe the purpose of a temperature sensor within the 3D printer.

.....
.....
.....
..... [2]

MARK SCHEME

7(b)	1 mark for each bullet point (max 2) • To prevent overheating // ensure material is hot enough • ...by identifying the temperature of the object (being printed) • ...by identifying the temperature of the material being used	2
------	--	----------

QUESTION

9 (a) Explain the importance of feedback in a control system.

.....

.....

.....

..... [2]

MARK SCHEME

9(a)	One mark for each bullet point (max 2) - Feedback ensures that a system operates within set criteria / constraints ...by enabling system output to affect (subsequent) system input ...thus allowing conditions to be <u>automatically</u> adjusted	2
------	---	----------

QUESTION

10 (a) Explain the importance of feedback in a control system.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

10(a)	1 mark for each bullet point: - to ensure the system operates with the given criteria ... by enabling system output to affect subsequent system input ... thus allowing conditions to be <u>automatically</u> adjusted	3
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QUESTION

- (b) (i) Identify **one** sensor that could be used in a car alarm system.

Justify your choice.

Sensor

Justification

.....

.....

.....

[2]

MARK SCHEME

10(b)(i)	1 mark for identification of a suitable sensor 1 mark for corresponding justification Example: • sound sensor • if a sound occurs inside the car the alarm is activated • infra-red sensor • senses the heat of person in the car / infra-red beams are broken • pressure sensor • an intruder sits in the driver's seat	2
----------	--	----------

QUESTION

- (c) The temperature in her refrigerator must be kept between 4 and 6 degrees Celsius.

The microprocessor in the refrigerator turns on the cooling if the temperature is too high, and turns off the cooling if the temperature is too low.

Explain why the system in the refrigerator is a control and not a monitoring system.

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.....

.....

.....

..... [2]

MARK SCHEME

5(c)	1 mark per bullet point • The system uses feedback • The system causes the temperature to change // produces an action	2
------	---	----------